



## Inventory in Ansible

### *Understanding Configuration Management & Automation*

#### **Introduction**

In Ansible, an **Inventory** is a list of **managed nodes (hosts)** that Ansible can connect to and automate. It helps define **groups of servers** for organized management.

This guide will cover:

- **What is an Inventory in Ansible?**
  - **Types of Inventories**
  - **How to create and use an Inventory**
  - **Managing dynamic inventories**
  - **Common issues and troubleshooting**
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#### ***What is an Inventory in Ansible?***

An **Inventory file** contains the **IP addresses or hostnames** of the machines Ansible will manage. It also defines **groups** of servers to apply specific tasks.

💡 *Think of an Inventory as a **contact list** for Ansible, allowing it to organize and communicate with remote servers!*

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#### ***Types of Inventories in Ansible***

| Inventory Type          | Description                              |
|-------------------------|------------------------------------------|
| <b>Static Inventory</b> | A manually created file listing servers. |



| Inventory Type           | Description                                                                     |
|--------------------------|---------------------------------------------------------------------------------|
| <b>Dynamic Inventory</b> | Automatically fetches server details from cloud providers like AWS, Azure, GCP. |

💡 Most beginners start with a **static inventory**, while advanced users use **dynamic inventories** for cloud automation.

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## Creating a Static Inventory File

### Step 1: Define an Inventory File (**inventory.ini**)

```
[webservers]
192.168.1.10
192.168.1.11

[databases]
db.example.com

[loadbalancers]
lb1.example.com
lb2.example.com
```

### Step 2: Test Connectivity

Check if Ansible can reach all hosts:

```
ansible all -m ping -i inventory.ini
```

💡 If you see **"pong"** messages, the connection is successful!

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## Advanced Inventory Features

### 1. Using Variables in Inventory

You can assign **custom variables** to specific hosts or groups.

Example (**inventory.ini**):

```
[webservers]
192.168.1.10 ansible_user=ubuntu ansible_port=22
192.168.1.11 ansible_user=root ansible_port=2222
```

Test connection:

```
ansible webservers -m ping -i inventory.ini
```

💡 *This allows you to use **different SSH users and ports** for different servers!*

### 2. Organizing Hosts into Groups

You can **combine groups** into larger categories.

Example (**inventory.ini**):

```
[webservers]
192.168.1.10
192.168.1.11

[databases]
db.example.com

[all_servers:children]
webservers
databases
```



Run a command on all servers in **all\_servers**:

```
ansible all_servers -m uptime -i inventory.ini
```

💡 *Grouping servers makes automation more efficient!*

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## ***Using a Dynamic Inventory (For Cloud Environments)***

A **Dynamic Inventory** fetches servers from cloud providers like AWS, Azure, and GCP **in real-time**.

### **Step 1: Install the AWS Inventory Plugin**

```
pip install boto3 botocore
```

### **Step 2: Use the AWS Inventory Script**

```
ansible-inventory --list -i aws_ec2.yaml
```

💡 *Dynamic inventories update automatically, reducing manual effort!*

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## ***Running Playbooks with Inventory***

Define an Inventory:

```
[webservers]
192.168.1.10
192.168.1.11
```

Create a Playbook (**webserver.yml**):

```
---
- name: Install Apache
  hosts: webservers
```



```
become: yes

tasks:

  - name: Install Apache Web Server

    apt:

      name: apache2

      state: present
```

Run the Playbook:

```
ansible-playbook -i inventory.ini webserver.yml
```

💡 *Inventories help Playbooks **apply configurations to specific groups of servers!***

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### ***Common Issues and Troubleshooting***

| Problem             | Solution                                                                   |
|---------------------|----------------------------------------------------------------------------|
| Host unreachable    | Check if SSH is enabled on the remote machine                              |
| Permission denied   | Use <code>ansible_user</code> and <code>ansible_become</code> in inventory |
| Connection timeout  | Verify the server's IP, firewall settings, and SSH access                  |
| Invalid YAML syntax | Ensure proper indentation in Playbooks                                     |

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## Know More (FAQs)

### 1. Can I have multiple inventory files?

Yes! You can pass multiple inventories using:

```
ansible-playbook -i inventory1.ini -i inventory2.ini playbook.yml
```

### 2. How do I list all available hosts in an inventory?

Run:

```
ansible-inventory --list -i inventory.ini
```

### 3. Can I use JSON/YAML instead of INI for inventory?

Yes! Example JSON Inventory:

```
{  
  "webservers": {  
    "hosts": ["192.168.1.10", "192.168.1.11"]  
  }  
}
```

### 4. How do I manage cloud servers with dynamic inventory?

Use cloud inventory plugins like **aws\_ec2**, **gcp\_compute**, or **azure\_rm**.

### 5. How do I check which groups a host belongs to?

Run:

```
ansible-inventory --graph -i inventory.ini
```

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## **Conclusion**

Ansible **Inventory** is a powerful tool for **organizing, managing, and automating servers** efficiently. It supports **static and dynamic inventories** for flexible automation.

👉 Try setting up your own **Inventory file** and automate server tasks!